

ADITHYA ABHIRAM CHITTIMALLA

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PROFILE:

Final Year B.Tech Computer Science student. A problem solving and backend engineering enthusiast. Designed and implemented production-style backend systems using Python, Flask, PostgreSQL, Redis, Celery, Docker, and AWS including asynchronous event processing and multi-service architectures Solved 400+ LeetCode problems, interested in backend, systems and cloud engineering.

TECHNICAL SKILLS:

Programming: C++, Python, C

Backend and Architecture: Flask, REST APIs, Celery, Microservice Patterns, MVC, JWT Authentication

Databases and Caching: Redis, PostgreSQL, SQLite

Cloud and DevOps: AWS (EC2, S3, RDS), Docker, Docker Compose, Git, GitHub, Pytest, Nginx

Web: HTML, CSS

CS Fundamentals: DSA, Operating Systems, Computer Architecture, OOP, Database Systems

Problem Solving: 400+ Problems (Focus on DSA, Arrays, Strings, Trees, Graphs, DP and Greedy)

PROJECTS:

AbyFlow – Async SaaS Billing API | Python, Flask, Stripe, Celery, Redis, PostgreSQL, Docker | [GitHub](#) | [Live Demo](#)

- Architected an event-driven billing API using Flask and PostgreSQL, integrating Stripe Checkout for secure subscription handling.
- Designed a non-blocking webhook system using Celery and Redis to process payment successes asynchronously, preventing API timeouts and duplicate database writes.
- Implemented robust authentication using JWT and Bcrypt, and containerized the multi-service architecture (Web API, Worker, DB, Redis) using Docker Compose.
- Deployed to AWS EC2 behind an Nginx reverse proxy with HTTPS, running alongside a second production service with isolated internal ports for the app, database, and cache layers.

AbyVest – Real-Time Portfolio Tracker and Data Aggregator | Python, Flask, Redis, PostgreSQL, Finnhub API, Docker | [GitHub](#) | [Live Demo](#)

- Engineered a stock portfolio web app integrating Finnhub and external LLM APIs, implementing a Redis caching layer (60s TTL for quotes, 24h for profiles) to reduce external API calls and prevent rate-limiting.
- Developed a service-layer architecture to decouple external API logic from route handlers, improving code maintainability and testability.
- Built a context-injection system that dynamically formats user portfolio data from PostgreSQL into structured prompts for AI analysis.
- Deployed to the same AWS EC2 host as AbyFlow via Nginx path-based routing, with PostgreSQL and Redis isolated on internal-only ports.

AbyNest – Modular Web Application Architecture | Python, Flask, SQLAlchemy, Cloudinary | [GitHub](#)

- Developed a server-rendered web application utilizing the Flask App Factory pattern and Blueprints to achieve strict MVC separation and modular routing.
- Implemented Role-Based Access Control (RBAC) ensuring users can only modify their own data, alongside one-to-many database relationships.
- Integrated third-party cloud storage (Cloudinary) for secure image uploads and implemented server-side form validation using WTForms.

ACADEMICS:

B.Tech Computer Science and Engineering

CVR College of Engineering | 2023-2027

CGPA : 8.5

CERTIFICATIONS & CLOUD TRAINING:

- AWS Academy Graduate - Cloud Operations | [Credly](#) | 2026
- AWS Academy Graduate - Cloud Foundations | [Credly](#) | 2026